

SalesGenie

AI Sales Assistant

Product Enquiries, Lead Qualification, Recommendations & Email

CAPSTONE PRODUCT & TECHNICAL SPECIFICATION

Updated Version — August 2026

Streamlit • LangChain • LangGraph • Gmail • Google Sheets • Langfuse

Document Overview

Purpose

Build a two-channel AI sales assistant for Oak & Ember Interiors. Customers can ask product questions through Streamlit or send a product enquiry by Gmail. Both channels use the same LangGraph recommendation workflow, and every enquiry is logged as a lead record for the sales team.

Current scope

- Streamlit chat with conversation context.
- Inbound Gmail product-enquiry processing.
- Customer need extraction and validation.
- Catalog retrieval using a local vector store.
- Deterministic filtering and ranking.
- LLM-generated catalog-grounded recommendations.
- Input and output guardrails.
- Automatic Gmail response to the original sender.
- Automatic lead logging to Google Sheets (name, email, enquiry, budget, recommended products, lead score, timestamp).
- Langfuse tracing for workflow steps, generations, latency, tokens and cost where available.
- Golden-dataset evaluation for correctness, relevance, hallucination/groundedness and related checks.

1. Project Summary & Example

Oak & Ember Interiors is an AI-powered furniture sales assistant that converts customer requirements into catalog-grounded product recommendations, while automatically capturing every enquiry as a qualified lead.

Example customer email

"Hi, I need an office chair under ₹50,000 with adjustable height and good back support."

System flow

- Receive the email.
- Run the input guardrail.
- Classify intent and extract needs.
- Retrieve relevant catalog products.
- Filter and rank products.
- Generate a recommendation using eligible catalog products only.
- Validate the recommendation.
- Log the lead (contact, requirement, budget, score, recommended products) to Google Sheets.
- Send the formatted recommendation directly back to the customer.

Example response

Hello,

Based on your requirement for an office chair under ₹50,000, here are the best matching products from the Oak & Ember Interiors catalog:

- Product P-014 — why it fits; catalog-supported trade-off; price.
- Product P-021 — why it fits; catalog-supported trade-off; price.

Please verify current price and availability.

Regards, Oak & Ember Interiors AI Sales Assistant

2. Current Sales Lead Identification Process — Challenges

Today, product enquiries are read and triaged manually before a lead is logged or a recommendation is sent. This creates the following bottlenecks:

- Time-consuming process to manually read and interpret emails.
- Manual entry of names and emails into Excel increases chances of mistakes, duplicates, or omissions.
- Recommending the product manually based on the enquiry and budget takes lots of time and is error-prone.
- No standard lead scoring; CRM logging is not real-time.

The AI Sales Assistant addresses each of these points: intent/need extraction removes manual reading, an automated write to Google Sheets replaces manual data entry, deterministic filtering plus an LLM recommender replaces manual product matching, and a rule-based lead score is written alongside each record so CRM logging happens in near real time.

3. Users, Goals & Success Metrics

Users

- Potential customer — receives product guidance based on requirements and budget.
- Project evaluator — verifies workflow quality, safety, evaluation and observability.
- Sales team / PM — reviews logged leads and scores in Google Sheets instead of manual CRM entry.

Success metrics

Metric	Target
Recommendation relevance	≥ 4/5
Groundedness / hallucination control	≥ 4.5/5
Hard-constraint compliance	100%
Critical hallucinations	0
Trace coverage	100% of evaluation runs
Gmail response correctness	100% for tested cases
Lead-log accuracy (Google Sheets)	100%, zero duplicate rows
Time-to-first-response (email received → reply sent)	< 5 minutes, vs. hours manually

4. Strategic Rationale

Q1. Agentic AI interventions in the Sales funnel

a) Lead intake & scoring agent

- **User & pain:** Sales admin spends hours reading enquiry emails and typing them into Excel.
- **Agentic idea:** An agent reads inbound Gmail messages, extracts contact/requirement/budget, applies a rule-based lead score, and writes a de-duplicated row to Google Sheets via the Sheets API.
- **Primary metric:** Manual data-entry time per lead; direction: decrease.
- **Assumptions & risk:** Assumes enquiries arrive in reasonably structured English; risk: malformed or spam emails inflate false leads if the input guardrail is weak.

b) Recommendation agent

- **User & pain:** Matching a customer's stated need and budget to the catalog by hand is slow and inconsistent between reps.
- **Agentic idea:** The existing LangGraph workflow retrieves, filters and ranks catalog products, then an LLM explains only the eligible, catalog-grounded shortlist.
- **Primary metric:** Recommendation relevance score; direction: increase toward the $\geq 4/5$ target.
- **Assumptions & risk:** Assumes the catalog is complete and current; risk: stale catalog data leads to recommending discontinued products.

c) Weekly sales-insight agent

- **User & pain:** PMs and sales leads have no standing view of enquiry volume, common requirements or conversion without pulling a manual report.
- **Agentic idea:** A scheduled agent reads the Google Sheet lead log, aggregates volume/category/budget trends and lead scores, and generates a short weekly summary.
- **Primary metric:** Time-to-insight (log $\rightarrow$ summary available); direction: decrease.
- **Assumptions & risk:** Assumes lead volume is high enough for trends to be meaningful; risk: small sample sizes make weekly summaries noisy early on.

Q2. Adoption stakeholders & benefits

Stakeholder	Decision / concern
Founder / Head of Sales	Owens go/no-go; wants faster lead response and pipeline visibility without adding headcount.
Sales / support reps	Direct users of the Sheet/CRM output; concerned about losing control over tone and accuracy of replies.
IT / Ops	Owens Gmail and Sheets access, API keys and data-retention practice.
AI Product Manager	Owens workflow design, guardrails, evaluation and rollout plan.
Customers	Indirect stakeholder; expects fast, accurate, non-spammy replies.

Org-wide benefits

- Faster, more consistent responses to product enquiries, improving conversion.
- A single, de-duplicated, real-time lead log instead of fragmented manual spreadsheets.
- Lower operating cost per enquiry as manual reading, entry and matching are automated.

PM-level benefits

- Full observability (Langfuse traces) into every workflow step, without manual QA.

- A golden-dataset evaluation loop that quantifies quality before and after changes.
- Weekly, data-backed insight into demand patterns to inform catalog and pricing decisions.

5. Solution Architecture

Entry points

- Streamlit Chat → Shared LangGraph Workflow → Response
- Gmail Inbox → Shared LangGraph Workflow → Response + Google Sheets lead log

Workflow

- Guardrail → Intent → Needs → Needs Validation
- Clarification / Insufficient response when required
- Retrieval → Filtering → Ranking
- Recommendation → Recommendation Validation → Response
- Lead logging → Google Sheets (Gmail channel)

Technology

- Python, Streamlit, LangGraph, LangChain, Pydantic
- Local vector store, SQLite/checkpointing
- Gmail API
- Google Sheets API
- Langfuse

6. Functional Requirements

Streamlit

- Accept product enquiries.
- Maintain conversation state.
- Ask clarification questions when needed.
- Retrieve, filter and rank products.
- Display grounded recommendations.

Inbound Gmail

- Read incoming product-enquiry emails.
- Extract sender, subject, body, message ID and thread ID.
- Ignore messages from the demo mailbox and already processed messages.
- Run the shared recommendation workflow.
- Send the validated recommendation directly to the original sender.
- Preserve the Gmail thread where applicable.

Lead logging (Google Sheets) — new

- Append one row per processed enquiry: timestamp, sender name, sender email, requirement summary, budget, recommended product IDs, lead score, message/thread ID.
- Use message ID as a de-duplication key so re-processing never creates a duplicate row.
- Write only after the recommendation has passed validation, so the sheet reflects what was actually sent.

Gmail response rule

There is no human approval step. The validated response is sent automatically, and the lead is logged automatically.

7. Product Catalog, Retrieval & Ranking

Catalog fields

- product_id, name, category, price, description
- features, ideal_for, limitations, product_url, active

Processing

- Embed product information.
- Retrieve top five semantically relevant products.
- Remove inactive products.
- Apply category, budget and required-feature filters.
- Rank eligible products.
- Pass only eligible shortlisted products to the recommendation LLM.

Rule

Python determines product eligibility. The LLM explains eligible products and must not invent catalog facts.

8. Guardrails & Validation

Input

- Prompt-injection detection.
- Unsafe-request detection.
- Off-topic detection.
- Supported handling of greetings and furniture sales requests.

Recommendation

- Product IDs must exist in the eligible set.
- Prices must match the catalog.
- Features and claims must be supported by catalog data.
- Hard budget and mandatory features are validated deterministically.

Gmail

- Prevent duplicate processing.
- Prevent replies to the demo mailbox itself.
- Use the original sender as recipient.
- Send only the validated recommendation response.
- Handle Gmail failures without false success.

Google Sheets (lead data) — new

- Log only fields needed for follow-up (name, email, requirement, budget, score); no sensitive personal data beyond what the customer volunteered.
- Restrict Sheet access to authorized sales/PM accounts.
- De-duplicate by message ID before every write; handle Sheets API failures without false success.

9. Langfuse Observability

Trace / span coverage

- Customer turn / evaluation execution.
- Guardrail result.
- Intent classification generation.
- Needs extraction generation.
- Retrieval and returned product IDs.
- Filtering and eligible product IDs.
- Ranking.
- Recommendation generation.
- Recommendation validation.
- Final response.
- Gmail processing/send activity.
- Google Sheets lead-log activity.

Metrics

- Latency.
- Model.
- Token usage when provided by the model/provider integration.
- Cost when provider pricing information is available.
- Evaluation scores.

Evaluation scores

- Correctness.
- Relevance.
- Hallucination / groundedness.
- Hard-constraint checks.
- Recommendation fit.

10. Golden Dataset & Evaluation

Dataset

Use approximately 25–30 golden queries covering normal product requests, missing information, no-match cases, hard constraints, prompt injection, unsafe/off-topic requests and clarification cases.

Evaluation process

- Run every golden query through the same graph used by the application.
- Capture response and relevant workflow state.
- Run deterministic evaluators.
- Run LLM-as-a-judge evaluators.
- Calculate per-query scores.
- Write individual scores to Langfuse.
- Calculate average scores across the golden dataset.
- Review failures and rerun after fixes.

Primary evaluators

- Correctness — requirements and facts handled correctly.
- Relevance — response directly addresses the request.
- Hallucination / groundedness — claims are supported by catalog data.
- Hard constraints — budget, required features and valid product IDs.
- Recommendation fit — selected products fit the stated need.

11. Gmail Workflow

Inbound flow

- Monitor the Oak & Ember Interiors Gmail inbox.
- Read new/unprocessed messages.
- Extract sender, subject and enquiry body.
- Run the shared LangGraph workflow.
- Generate and validate the recommendation.
- Append the lead record to Google Sheets.
- Create a properly formatted Gmail reply.
- Send the reply directly to the customer.

Gmail response format

- Greeting.
- Short summary of the customer's requirement.
- Recommended products with product ID and price.
- Why each product fits.
- Trade-off / limitation where available.
- Approved product link.
- Price/availability verification note.
- Thanks / closing message.

12. SalesGenie — Design Rationale

SalesGenie is the Agentic AI Sales Assistant answer to Q3: it qualifies leads, sends product recommendations, and produces weekly sales insight, built on the same shared LangGraph workflow described in this specification.

Approach, tools and data structures

- Reuse the existing guardrail → intent → needs → retrieval → filter → rank → recommend → validate pipeline as the core reasoning path; add a lead-scoring node and a Sheets-logging node after validation.
- Lead score is a simple deterministic rule (budget fit + required-feature match + response completeness), not an LLM judgment, so it stays auditable and repeatable.
- Tools/APIs: Gmail API (read/send), Google Sheets API (append/read), local vector store (retrieval), Langfuse SDK (tracing/eval).
- Data structures: a Lead record (contact, requirement, budget, score, recommended product IDs, timestamp, message ID) and the existing Product catalog schema; both kept as flat, auditable rows rather than nested objects, since the Sheet is the source of truth for the sales team.
- Weekly insight job reads the Sheet (not the LLM's memory) and asks the LLM to summarize only aggregated, pre-computed statistics — keeping the summary grounded in real data rather than model recall.

Success metrics & baselining

Metric	Baseline method
Lead qualification accuracy	Compare agent's lead score against a manually scored sample of the golden dataset.
Recommendation relevance & groundedness	LLM-as-judge and deterministic checks against the same golden dataset used today.
Time-to-first-response	Compare current manual turnaround (hours) against automated turnaround (minutes) on a sample of enquiries.
Lead-log accuracy & duplication rate	Audit a sample of Google Sheet rows against source emails.
Weekly insight usefulness	PM/sales-rep review of the first four weekly summaries against actual pipeline activity.

Guardrails

- Privacy: log only customer-volunteered contact and requirement details; restrict Sheet access; no storage of payment or unrelated personal data.
- Grounding: recommendations and weekly insights are restricted to catalog data and Sheet aggregates respectively — the LLM never fabricates product facts or numbers.
- Fallbacks: on Gmail or Sheets API failure, the workflow retries once, then surfaces a failure trace in Langfuse rather than reporting false success; on low-confidence intent, the customer is asked a clarifying question instead of guessing.

Dummy data & assumptions

- Assumes a realistic dummy catalog (~30–50 SKUs across chairs, tables, storage, lighting) with the fields listed in Section 7.
- Assumes a dummy Gmail inbox and a dummy CRM export used only to validate the Sheets schema and de-duplication logic.
- Assumes enquiry volume is low enough in the demo (dozens, not thousands) that a Google Sheet is an adequate CRM substitute; a real CRM integration is a future extension, not in current scope.

Deliverables

- A working demo showing Streamlit chat, an inbound Gmail enquiry, the automatic reply, and the resulting Google Sheet row.
- A concise slide deck summarizing the problem, approach and outcomes.
- A short (~5 minute) demo video walking through the system and the design choices above.

13. Testing & Acceptance

Minimum tests

- Budget filtering.
- Category and feature filtering.
- Intent classification.
- Needs extraction.
- Recommendation validation.
- Prompt injection and unsafe requests.
- No-match handling.
- Gmail receive/process/reply.
- Duplicate Gmail message prevention.
- Google Sheets lead logging and de-duplication.
- Langfuse trace creation.
- Golden-dataset evaluation.

Acceptance checklist

- Streamlit and Gmail use the shared recommendation workflow.
- Clarification occurs when required information is missing.
- Every recommended product passes hard filters.
- Recommendations contain product ID, rationale and trade-off where applicable.
- Unsupported or unsafe requests are handled safely.
- Gmail replies go to the correct original sender.
- Every processed enquiry produces exactly one Google Sheets row.
- No human approval is required.
- Langfuse contains workflow traces and evaluation scores.
- Golden-dataset results are available for review.

14. Two-Week Implementation Plan

Day	Focus
1	Project setup, Gmail OAuth, Google Sheets API setup, Langfuse and catalog
2	Catalog ingestion and retrieval
3	Needs extraction, filtering and ranking
4	Recommendation workflow and validation
5	Streamlit chat
6	Inbound Gmail integration
7	Automatic Gmail response, formatting and lead scoring
8	Google Sheets lead logging, guardrails and Langfuse tracing
9	Golden dataset and evaluators
10	End-to-end testing and presentation

15. Project Structure

Folders

- `__pycache__` — Python bytecode cache
- `data` — catalog and supporting data files
- `database` — local vector store / checkpointing
- `evals` — evaluation artifacts and results

Root files

- `.env` — environment variables and secrets
- `aichains` — LLM chains
- `Alfiltering` — deterministic filters
- `Alguardrail` — guardrail logic
- `Alrecommender` — ranking/recommendation
- `Alretrieval` — catalog retrieval
- `app` — Streamlit entry point
- `config` — app configuration
- `evaluators` — evaluation functions
- `gmail_credentials` — Gmail OAuth credentials
- `gmail_processor` — Gmail message processing
- `gmail_services` — Gmail API integration
- `gmail_token` — Gmail OAuth token
- `golden_dataset` — golden test cases
- `google_service_account` — Google service-account credentials
- `google_sheets` — Google Sheets lead-logging integration
- `graph` — LangGraph definition
- `guardrail_nodes` — guardrail workflow nodes
- `ingest_catalog` — catalog ingestion into the vector store
- `langfuse_config` — Langfuse client/configuration
- `ReadMe` — project documentation
- `recommendation_nodes` — recommendation workflow nodes
- `requirements` — Python dependencies
- `response_nodes` — final response nodes
- `run_evals` — evaluation runner
- `sales_agent` — sales agent orchestration
- `sales_nodes` — sales workflow nodes
- `start_app` — application startup script
- `state` — workflow state
- `utils` — shared helper functions

Current implementation boundary

The project contains two entry channels, one shared recommendation workflow, automatic Gmail responses, automatic Google Sheets lead logging, guardrails, Langfuse observability and offline evaluation. Human approval and follow-up automation are outside the current scope.

16. Definition of Capstone Complete

The project is complete when a customer can enter through Streamlit chat or inbound Gmail and receive catalog-grounded recommendations that honor the stated requirements, with every enquiry captured as a lead.

- The shared LangGraph workflow works for both channels.
- Clarification, no-match and blocked-request paths work.
- Product eligibility is determined by deterministic rules.
- Recommendations are validated against the catalog.
- Inbound Gmail enquiries receive a formatted response automatically.
- Every inbound Gmail enquiry is logged to Google Sheets exactly once.
- Guardrail results are captured.
- Langfuse traces show important workflow steps and model activity.
- Golden-dataset evaluations produce per-query and aggregate scores.
- The project handles LLM, retrieval, Gmail and Sheets failures without false success.

Final deliverables

- Runnable Python repository.
- Product catalog and vector index.
- Streamlit chatbot.
- Gmail integration.
- Google Sheets lead log.
- Shared LangGraph workflow.
- Guardrails.
- Langfuse observability.
- Golden dataset and evaluation runner.
- Test results and README.